

Multi-Task Instruction Tuning

Training simultaneously on 60+ task types — how FLAN built a general instruction-following model

10 min

Duration

B3 – Apply

Bloom's

L3 – Advanced

Level

AI-FT-IN-TH-000013

ID

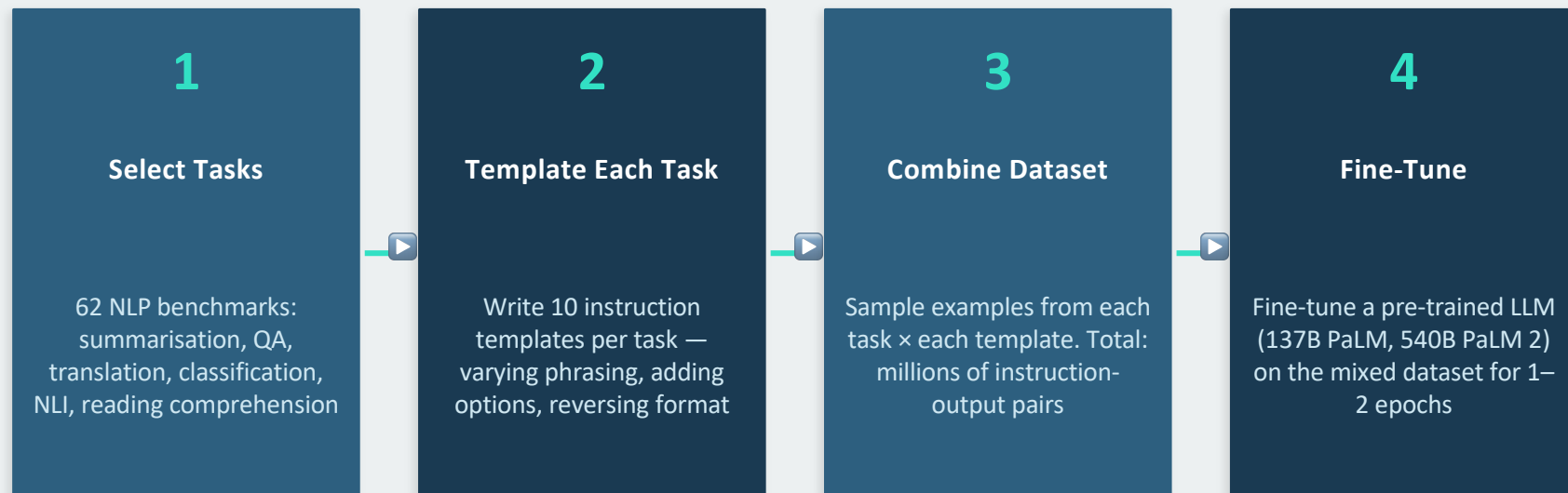
Multi-Task Instruction Tuning — Core Idea

Definition

Multi-task instruction tuning trains the model simultaneously on a mixture of many different NLP tasks, each reformulated as a natural language instruction. The model learns to generalise across tasks — improving zero-shot performance on unseen tasks.

- FLAN (Wei et al., 2022): 62 NLP tasks → 62 instruction templates each → the model learns task-general instruction following
- The diversity of tasks is more important than the volume of any single task
- Key finding: models tuned on task mixtures generalise better to held-out tasks than models tuned on individual tasks
- The instruction template variety per task teaches robustness to prompt phrasing

How FLAN Was Built — 4-Stage Process



Multi-Task vs Single-Task Fine-Tuning — When to Use Each

Scenario	Multi-Task IT	Single-Task FT
Goal: general assistant	✓ Best choice — broad task coverage	✗ Narrow capability
Goal: one specific domain	✗ Wasteful; dilutes task signal	✓ Best choice
Zero-shot generalisation to new tasks	✓ Multi-task gives strong zero-shot	✗ Doesn't generalise
Data: many task types available	✓ Leverage diversity	N/A
Data: single domain only	✗ Not enough task variety	✓ Focused training
Compute budget	High (many tasks × many templates)	Low (focused dataset)

KEY REFERENCES

- Wei, J. et al. (2021). Finetuned Language Models Are Zero-Shot Learners (FLAN). ICLR 2022.
- Chung, H.W. et al. (2022). Scaling Instruction-Finetuned Language Models (FLAN-T5). arXiv.
- Longpre, S. et al. (2023). The Flan Collection: Designing Data and Methods for Effective Instruction Tuning. ICML 2023.

 **Next: Concept 17: Data Quality vs Quantity (AI-FT-IN-TH-000014)**